

COL1000: Lab 1

Semester 2, 2025–26

Question 0:

Run IDLE on your computer. Type the following program

```
i=input('provide your choice a/b:')
if (i=="a"):
    print("*****\n*****\n***\n**\n*")
elif (i=="b"):
    print("*****\n ***** \n  ***  \n   *   ")
    print("   *   \n  ***  \n ***** \n*****")
else:
    print("invalid input")
```

Save the program to a file called “lab1.py”. Run the program.

Question 1: Pattern Printing

Write a Python program that takes as input either “a”, “b” or “c”, and depending on the input, it prints one of the following patterns. For any other input, it must print “Invalid Input”.

(a) Inverted left-aligned triangle

```
*****
****
***
**
*
```

(b) Hourglass (stars)

```
*****
*****
*****
***
*
***
```

```

*****
*****
*****

```

(c) Butterfly pattern

```

*      *
**     **
***    ***
*****
*****
***    ***
**     **
*      *

```

Question 2: Quadratic Equation Roots

Write a program that takes three real numbers **a**, **b**, and **c** as input and determines how many **real roots** the quadratic equation

$$ax^2 + bx + c = 0$$

has. You may assume that **a** is non-zero.

Question 3: Triangle Validity

Write a program that takes three positive integers **a**, **b**, and **c** as input and checks whether they can form a valid triangle. A triangle is valid if:

$$a + b > c, \quad b + c > a, \quad a + c > b$$

Print Valid Triangle or Invalid Triangle.